

Tower Of Hanoi – An Introduction

The tower of Hanoi Problem was believed to have recreated by a french mathematician, Edouard Lucas in 1883 , and published it in booklet named `Recreation mathematiques` in 1889.

The tower of Hanoi also called `The problem of Benaras Temple or Tower of Brahma`. According to a legend, the priests of Kasi Viswanath temple in India were asked by Brahma , the creator, to move 64 disks from first peg, of the three pegs, to the third peg. Even if it takes 1s to move a disk to another 1st peg, then it would take $2^{64} - 1$ or more than 580 billion years to complete the task.

The Mathematical Puzzle

The tower of hanoi is a mathematical puzzle , which consists of three rods. Initially there are `n` disks in the first rod such that the disk `i` is above the disk `j` , the radius of `j` greater than that of `i` .

Objective :

The objective is to transfer the disks into the last rod, such that at no point in times a larger disk is above a smaller disk. Moreover , only one disk can move at a time.

So, there are `n` disks 1, 2, 3, ... (2nd Disk above 1st Disk and 3rd Disk above 2nd Disk) having radii $r_1, r_2, \dots, r_n$ in the first rod, such that $r_1 > r_2 > r_3$ and so on, then at the end there should be `n` disks `1`, `2`, `3`, ... (2nd Disk above 1st Disk and 3rd Disk above 2nd Disk) having radii $r_1, r_2, \dots, r_n$, in the last rod, such that $r_1 > r_2 > r_3$ and so on.
